

Chapter 2

Descriptive Statistics

Elementary Statistics

Picturing the World

EIGHTH EDITION

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GLOBAL
EDITION



Chapter Outline

- 2.1 Frequency Distributions and Their Graphs
- 2.2 More Graphs and Displays
- 2.3 Measures of Central Tendency
- 2.4 Measures of Variation
- 2.5 Measures of Position

Section 2.1

Frequency Distributions and Their Graphs

Section 2.1 Objectives

- How to construct a frequency distribution including limits, midpoints, relative frequencies, cumulative frequencies, and boundaries
- How to construct frequency histograms, frequency polygons, relative frequency histograms, and ogives

Frequency Distribution

Frequency Distribution

- A table that shows **classes** or **intervals** of data with a count of the number of entries in each class.
- The **frequency, f** , of a class is the number of data entries in the class.

Class	Frequency, f
1 – 5	5
6 – 10	8
11 – 15	6
16 – 20	8
21 – 25	5
26 – 30	4

Frequency Distribution

- Each class has a **lower class limit**, which is the least number that can belong to the class, and an
- **upper class limit**, which is the greatest number that can belong to the class.

Class	Frequency, f
1 – 5	5
6 – 10	8
11 – 15	6
16 – 20	8
21 – 25	5
26 – 30	4

Lower class
limits

Upper class
limits

Frequency Distribution

- The **class width** is the distance between lower (or upper) limits of consecutive classes.
- The difference between the maximum and minimum data entries is called the **range**.

Class width
 $6 - 1 = 5$ ←

Class	Frequency, f
1 – 5	5
6 – 10	8
11 – 15	6
16 – 20	8
21 – 25	5
26 – 30	4

Constructing a Frequency Distribution

1. Decide on the number of classes.
 - Usually between 5 and 20; otherwise, it may be difficult to detect any patterns.
2. Find the class width.
 - Determine the range of the data.
 - Divide the range by the number of classes.
 - *Round up to the next convenient number.*

Constructing a Frequency Distribution

3. Find the class limits.
 - You can use the minimum data entry as the lower limit of the first class.
 - Find the remaining lower limits (add the class width to the lower limit of the preceding class).
 - Find the upper limit of the first class. Remember that classes cannot overlap.
 - Find the remaining upper class limits.

Constructing a Frequency Distribution

4. Make a tally mark for each data entry in the row of the appropriate class.
5. Count the tally marks to find the total frequency f for each class.

Example: Constructing a Frequency Distribution

The data set lists the cell phone screen times (in minutes) for 30 U.S. adults on a recent day. Construct a frequency distribution that has seven classes.

200	239	155	252	384	165	296	405	303	400
307	241	256	315	330	317	352	266	276	345
238	306	290	271	345	312	293	195	168	342

Solution: Constructing a Frequency Distribution

200 239 155 252 384 165 296 405 303 400
307 241 256 315 330 317 352 266 276 345
238 306 290 271 345 312 293 195 168 342

1. Number of classes = 7 (given)
2. Find the class width

$$\frac{\text{Range}}{\# \text{ classes}} = \frac{\text{max} - \text{min}}{\# \text{ classes}} = \frac{405 - 155}{7} = \frac{250}{7} \approx 35.71$$

Round up to 36

Solution: Constructing a Frequency Distribution

3. Use 155 (minimum value) as first lower limit. Add the class width of 36 to get the lower limit of the next class.

$$155 + 36 = 191$$

Find the remaining lower limits.

Class
width = 36

Lower limit	Upper limit
155	190
191	226
227	262
263	298
299	334
335	370
371	406

Solution: Constructing a Frequency Distribution

The upper limit of the first class is 190 (one less than the lower limit of the second class).

Add the class width of 36 to get the upper limit of the next class.

$$190 + 36 = 226$$

Find the remaining upper limits.

Lower limit	Upper limit
155	190
191	226
227	262
263	298
299	334
335	370
371	406

Class
width = 36

Solution: Constructing a Frequency Distribution

4. Make a tally mark for each data entry in the row of the appropriate class.
5. Count the tally marks to find the total frequency f for each class.

The diagram illustrates the construction of a frequency distribution table. It features a table with three columns: Class, Tally, and Frequency, f . The rows represent different expense classes, with tally marks and corresponding frequencies. Annotations include: 'Expenses' pointing to the Class column, 'Number of adults' pointing to the Frequency column, and a note 'Check that the sum of the frequencies equals the number in the sample.' pointing to the total frequency $\Sigma f = 30$.

Class	Tally	Frequency, f
155–190		3
191–226		2
227–262		5
263–298		6
299–334		7
335–370		4
371–406		3
		$\Sigma f = 30$

Determining the Midpoint

Midpoint of a class

$$\frac{(\text{Lower class limit}) + (\text{Upper class limit})}{2}$$

$$\text{Midpoint} = \frac{155 + 190}{2} = 172.5$$

Midpoint of first class.

Using the class width of 36, the remaining mid-points are

$$172.5 + 36 = 208.5$$

Midpoint of second class


$$208.5 + 36 = 244.5$$

Midpoint of third class

and so on.

Determining the Relative Frequency

Relative Frequency of a class

- Portion or percentage of the data that falls in a particular class.
- $$\text{relative frequency} = \frac{\text{class frequency}}{\text{Sample size}} = \frac{f}{n}$$

Note that $n = \sum f$.

Determining the Cumulative Frequency

Cumulative frequency of a class

- The sum of the frequency for that class and all previous classes.
- The cumulative frequency of the last class is equal to the sample size n .

Example: Finding Midpoints, Relative and Cumulative Frequencies

Using the frequency distribution in previous example, find the midpoint, relative frequency, and cumulative frequency of each class. Describe any patterns.

Solution: Finding Midpoints, Relative and Cumulative Frequencies

- The midpoints, relative frequencies, and cumulative frequencies of the first five classes are calculated as follows:

Class	f	Midpoint	Relative frequency	Cumulative frequency
155–190	3	$\frac{155 + 190}{2} = 172.5$	$\frac{3}{30} = 0.1$	3
191–226	2	$\frac{191 + 226}{2} = 208.5$	$\frac{2}{30} \approx 0.07$	$3 + 2 = 5$
227–262	5	$\frac{227 + 262}{2} = 244.5$	$\frac{5}{30} \approx 0.17$	$5 + 5 = 10$
263–298	6	$\frac{263 + 298}{2} = 280.5$	$\frac{6}{30} = 0.2$	$10 + 6 = 16$
299–334	7	$\frac{299 + 334}{2} = 316.5$	$\frac{7}{30} \approx 0.23$	$16 + 7 = 23$

Solution: Finding Midpoints, Relative and Cumulative Frequencies

- The remaining midpoints, relative frequencies, and cumulative frequencies are shown in the expanded frequency distribution below.

**Frequency Distribution for
Cell Phone Screen Times (in minutes)**

Number of adults	Class	Frequency, f	Midpoint	Relative frequency	Cumulative frequency	Portion of adults
	155–190	3	172.5	0.1	3	
	191–226	2	208.5	0.07	5	
	227–262	5	244.5	0.17	10	
	263–298	6	280.5	0.2	16	
	299–334	7	316.5	0.23	23	
	335–370	4	352.5	0.13	27	
	371–406	3	388.5	0.1	30	
		$\Sigma f = 30$		$\Sigma \frac{f}{n} = 1$		

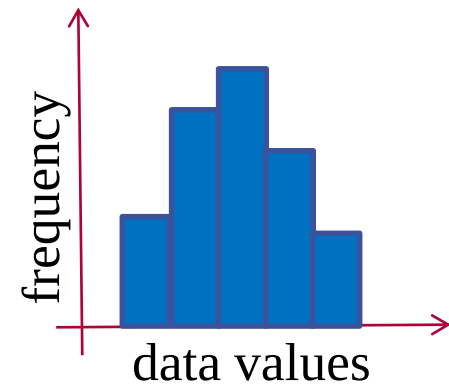
Solution: Finding Midpoints, Relative and Cumulative Frequencies

- There are several patterns in the data set. For instance, the most common range for the time is 299 to 334 minutes. Also, about half of the times are less than 299 minutes.

Graphs of Frequency Distributions

Frequency Histogram

- A bar graph that represents the frequency distribution.
- The horizontal scale is quantitative and measures the data values.
- The vertical scale measures the frequencies of the classes.
- Consecutive bars must touch.



Class Boundaries

Class boundaries

- Because consecutive bars of a histogram must touch, bars must begin and end at class boundaries instead of class limits.
- The numbers that separate classes *without* forming gaps between them.

Example: Constructing a Frequency Histogram

Draw a frequency histogram for the frequency distribution in the previous example. Describe any patterns.

**Frequency Distribution for
Cell Phone Screen Times (in minutes)**

Times Number of adults	Class	Frequency, f	Midpoint	Relative frequency	Cumulative frequency	Portion of adults
	155–190	3	172.5	0.1	3	
	191–226	2	208.5	0.07	5	
	227–262	5	244.5	0.17	10	
	263–298	6	280.5	0.2	16	
	299–334	7	316.5	0.23	23	
	335–370	4	352.5	0.13	27	
	371–406	3	388.5	0.1	30	
		$\Sigma f = 30$		$\Sigma \frac{f}{n} = 1$		

Solution: Constructing a Frequency Histogram

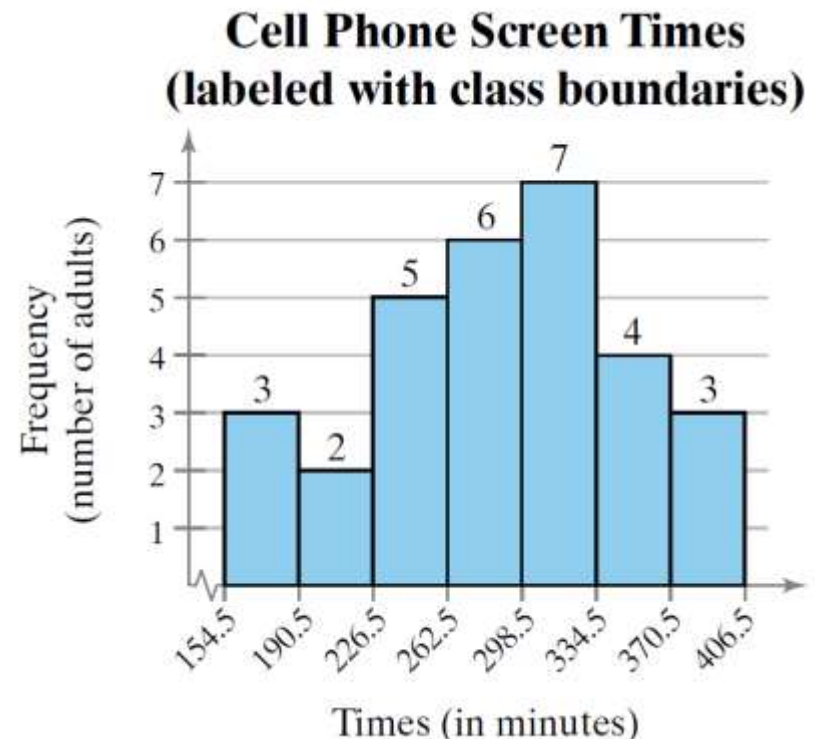
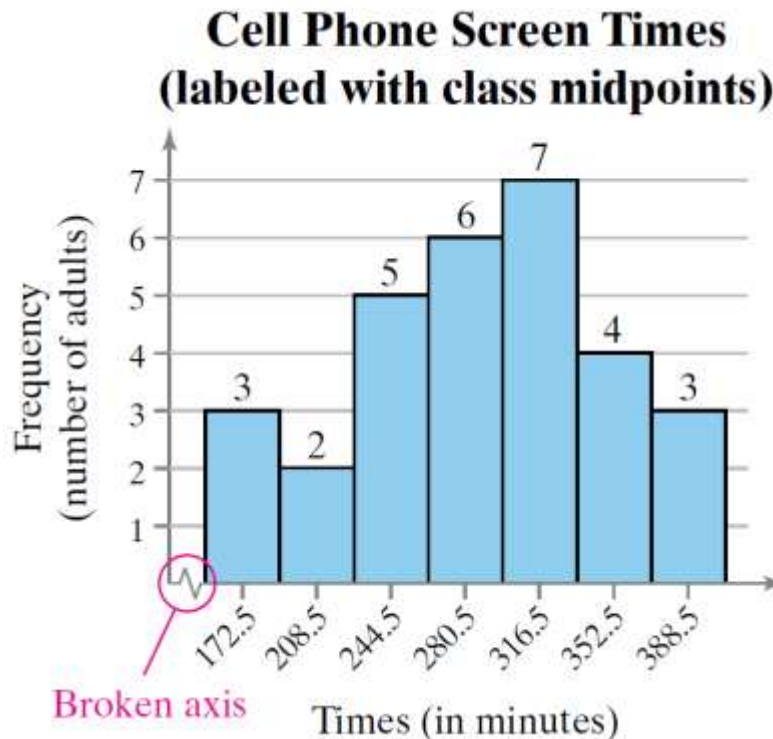
- First, find the class boundaries
- The distance from the upper limit of the first class to the lower limit of the second class is $191 - 190 = 1$.
- Half this distance is 0.5.

Class	Class boundaries	Frequency, f
155–190	154.5–190.5	3
191–226	190.5–226.5	2
227–262	226.5–262.5	5
263–298	262.5–298.5	6
299–334	298.5–334.5	7
335–370	334.5–370.5	4
371–406	370.5–406.5	3

- First class lower boundary = $155 - 0.5 = 154.5$
- First class upper boundary = $190 + 0.5 = 190.5$

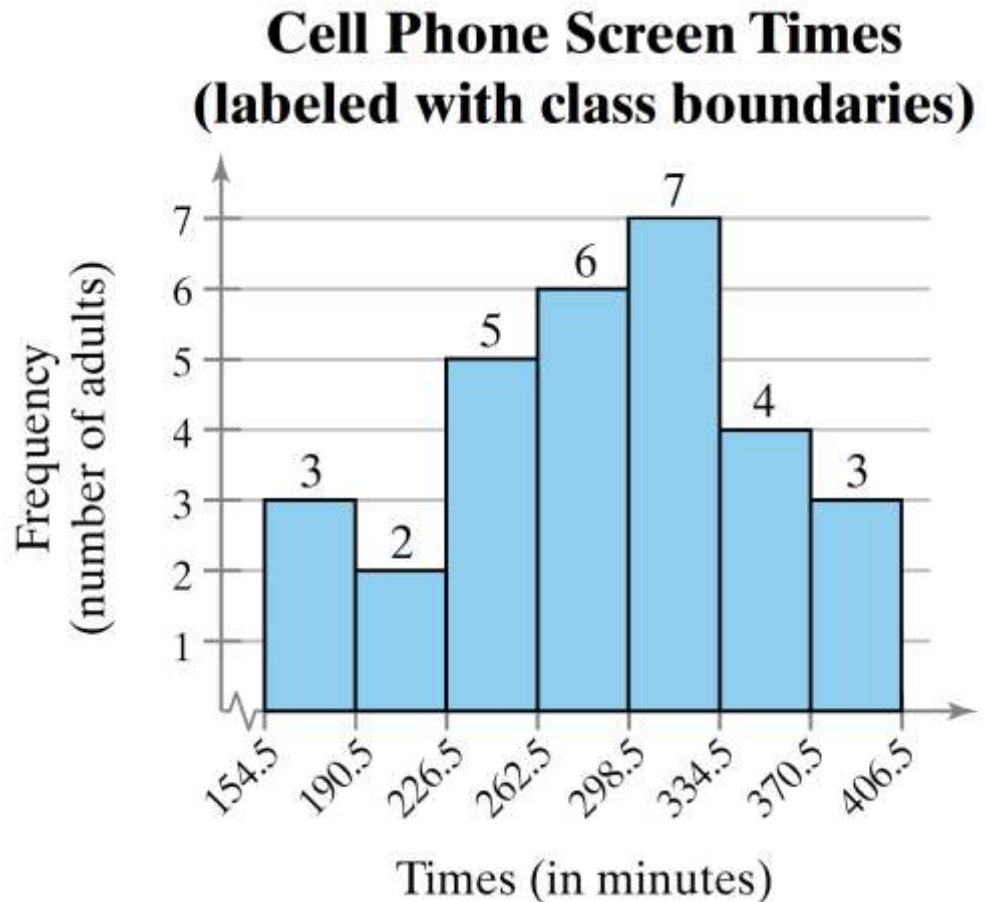
Solution: Constructing a Frequency Histogram

You can mark the horizontal scale either at the midpoints or at the class boundaries. Both histograms are shown below.



Solution: Constructing a Frequency Histogram

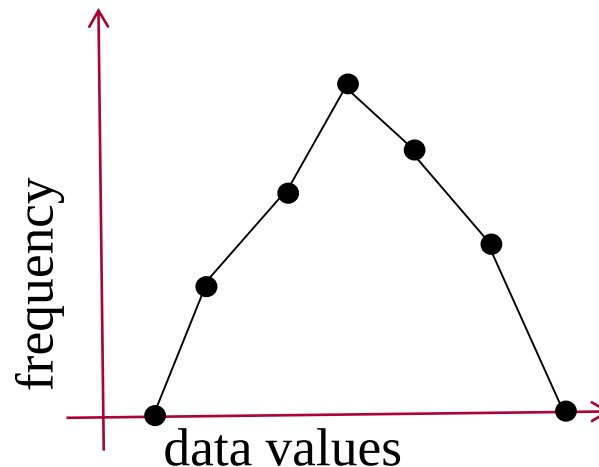
You can determine that two-thirds of the adults are spending more than 262.5 minutes each day using their cell phones.



Graphs of Frequency Distributions

Frequency Polygon

- A line graph that emphasizes the continuous change in frequencies.



Example: Constructing a Frequency Polygon

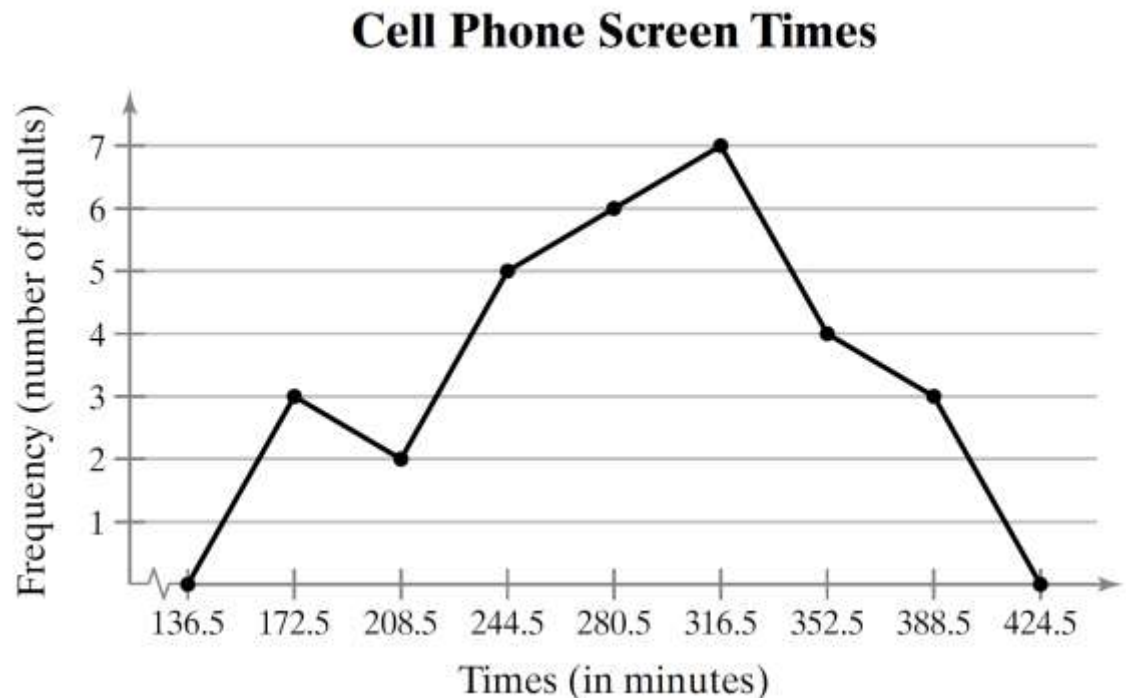
Draw a frequency polygon for the frequency distribution in previous example. Describe any patterns.

Class	Class boundaries	Frequency, f
155–190	154.5–190.5	3
191–226	190.5–226.5	2
227–262	226.5–262.5	5
263–298	262.5–298.5	6
299–334	298.5–334.5	7
335–370	334.5–370.5	4
371–406	370.5–406.5	3

Solution: Constructing a Frequency Polygon

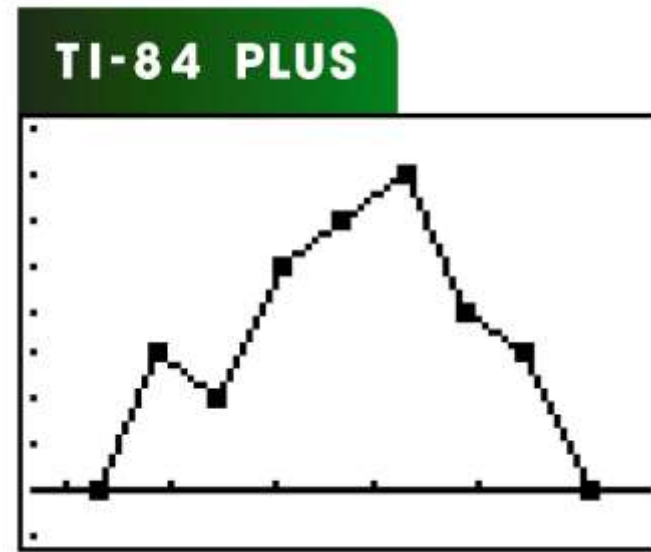
To construct the frequency polygon, use the same horizontal and vertical scales that were used in the histogram labeled with class midpoints.

The graph should begin and end on the horizontal axis, so extend the left side to one class width before the first class midpoint and extend the right side to one class width after the last class midpoint.



Solution: Constructing a Frequency Polygon

You can check your answer using technology.

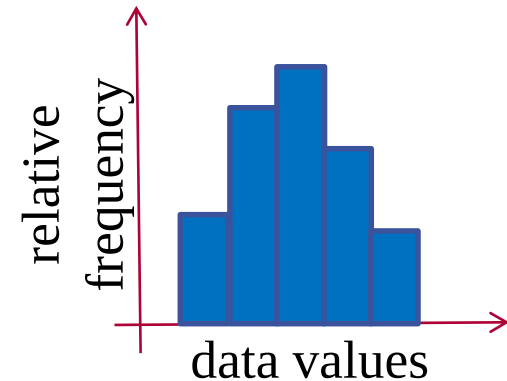


You can see that the frequency of adults increases up to time of 316.5 minutes and then the frequency decreases.

Graphs of Frequency Distributions

Relative Frequency Histogram

- Has the same shape and the same horizontal scale as the corresponding frequency histogram.
- The vertical scale measures the **relative frequencies**, not frequencies.

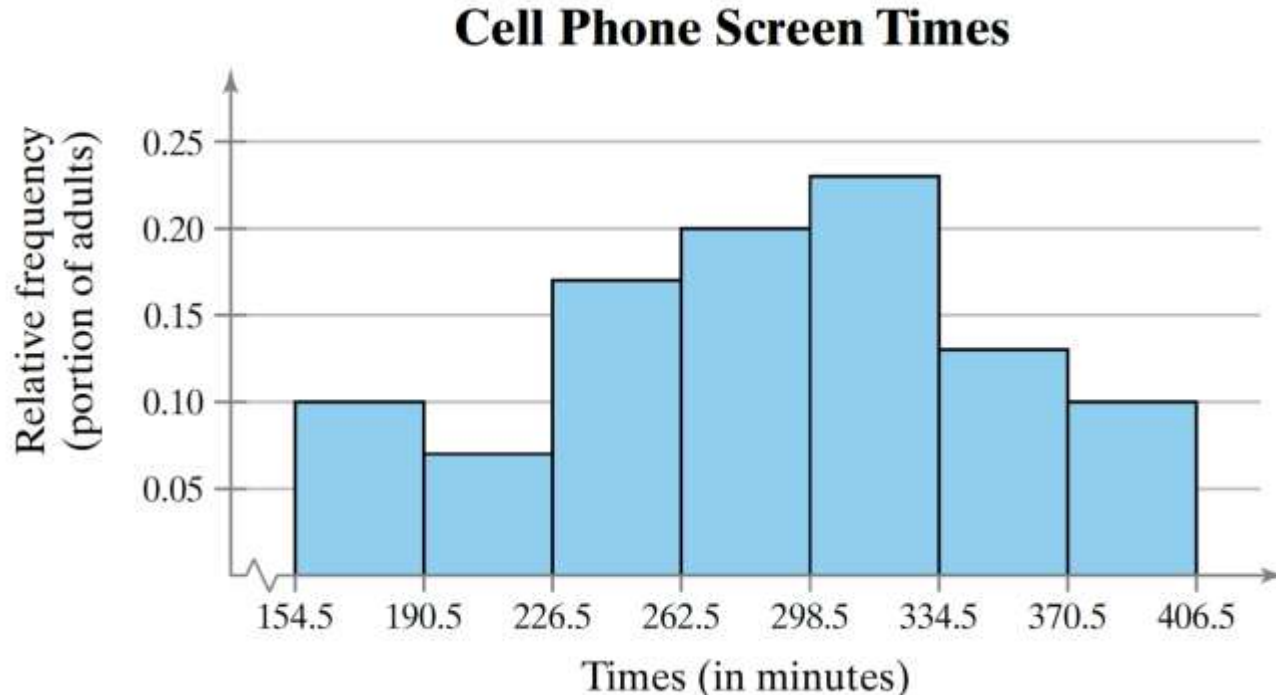


Example: Constructing a Relative Frequency Histogram

Construct a relative frequency histogram for the second example.

Class	Class boundaries	Frequency, f
155–190	154.5–190.5	3
191–226	190.5–226.5	2
227–262	226.5–262.5	5
263–298	262.5–298.5	6
299–334	298.5–334.5	7
335–370	334.5–370.5	4
371–406	370.5–406.5	3

Solution: Constructing a Relative Frequency Histogram

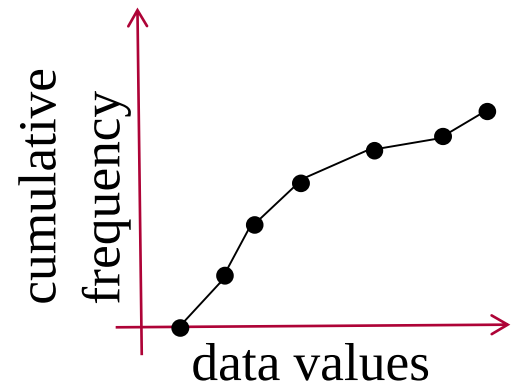


From this graph, you can quickly see that 0.2, or 20%, of the adults screen times between 262.5 and 298.5 minutes.

Graphs of Frequency Distributions

Cumulative Frequency Graph or Ogive

- A line graph that displays the cumulative frequency of each class at its upper class boundary.
- The upper boundaries are marked on the horizontal axis.
- The cumulative frequencies are marked on the vertical axis.



Constructing an Ogive

1. Construct a frequency distribution that includes cumulative frequencies as one of the columns.
2. Specify the horizontal and vertical scales.
 - The horizontal scale consists of the upper class boundaries.
 - The vertical scale measures cumulative frequencies.
3. Plot points that represent the upper class boundaries and their corresponding cumulative frequencies.

Constructing an Ogive

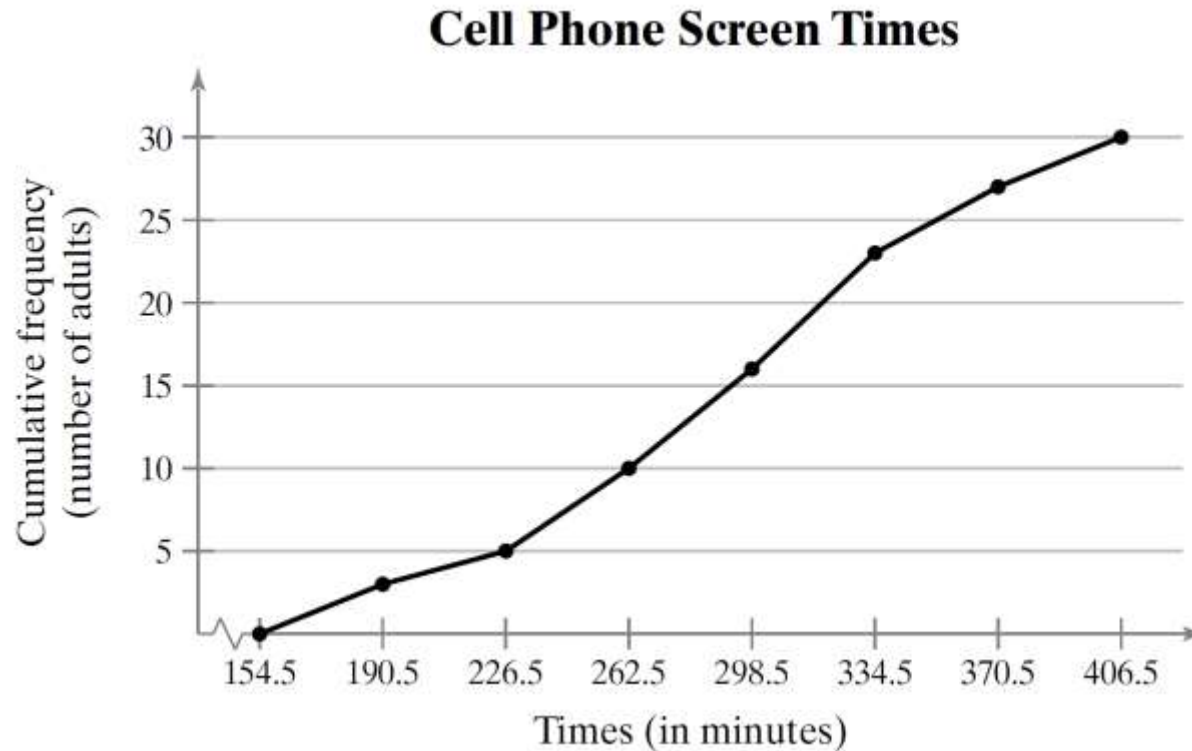
4. Connect the points in order from left to right.
5. The graph should start at the lower boundary of the first class (cumulative frequency is zero) and should end at the upper boundary of the last class (cumulative frequency is equal to the sample size).

Example: Constructing an Ogive

Construct an ogive for the second example frequency distribution.

Upper class boundary	f	Cumulative frequency
190.5	3	3
226.5	2	5
262.5	5	10
298.5	6	16
334.5	7	23
370.5	4	27
406.5	3	30

Solution: Constructing an Ogive



From the ogive, you can see that 10 adults had screen times of 262.5 minutes or less. Also, the greatest increase in cumulative frequency occurs between 298.5 minutes and 334.5 minutes.

Example: Technology to Construct Histograms

Use technology to construct a histogram for the frequency distribution in the second example.

Here are instructions for creating a histogram on a TI-84 Plus.

STAT ENTER

Enter midpoints in L1.

Enter frequencies in L2.

2nd STAT PLOT

Turn on Plot 1.

Highlight Histogram.

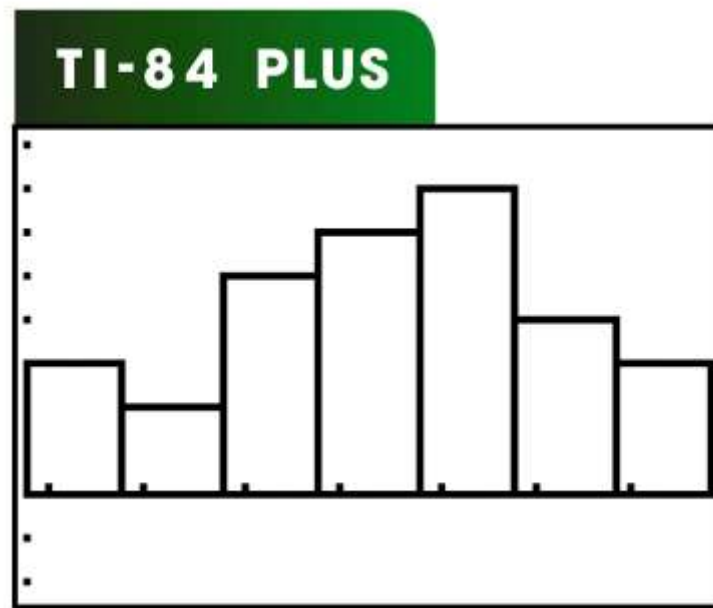
Xlist: L1

Freq: L2

ZOOM 9

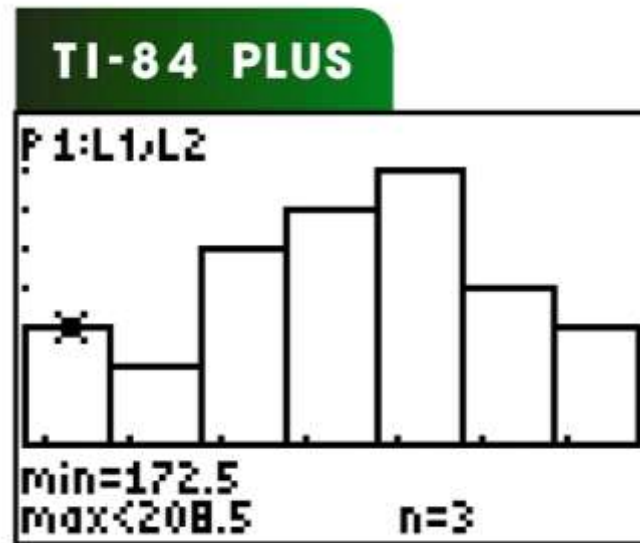
Example: Technology to Construct Histograms

- Using the instructions for a TI-84 Plus shown, you can draw a histogram similar to the one below.



Example: Technology to Construct Histograms

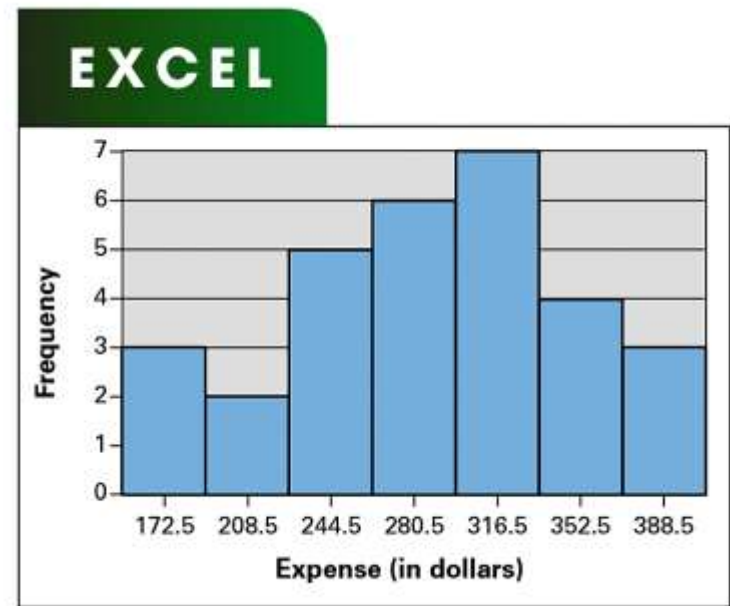
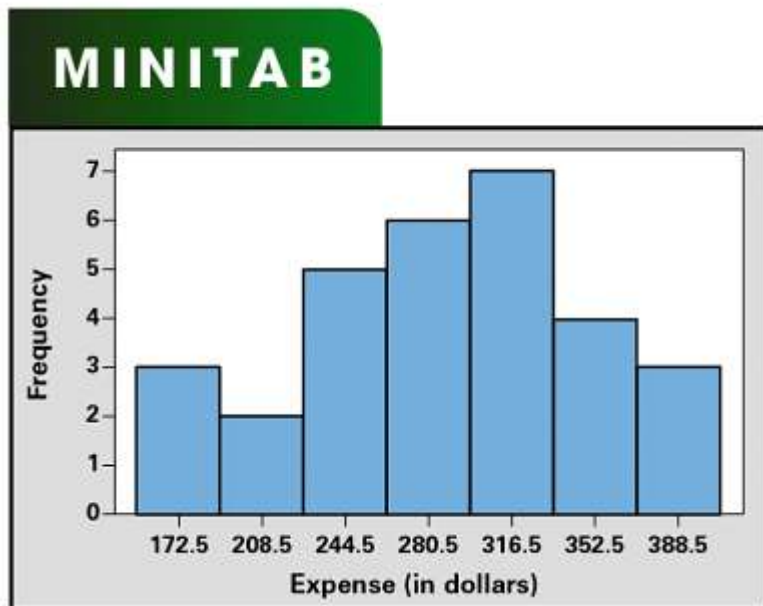
- After pressing TRACE, the midpoint and the frequency of the first class are displayed, as shown in the figure.



- Use the right and left arrow keys to move through each bar.

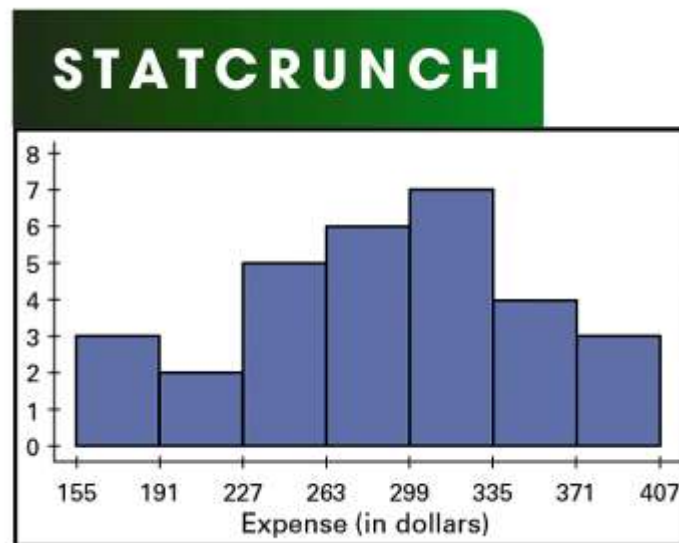
Example: Technology to Construct Histograms

- Histograms made using Minitab and Excel are shown below.



Example: Technology to Construct Histograms

- Histograms made using StatCrunch is shown below.



Exercise 1

Use the given frequency distribution to find the

- a. class width.
- b. class midpoints.
- c. class boundaries.

Cleveland, OH
High Temperature (°F)

Class	Frequency, f
20 – 30	19
31 – 41	43
42 – 52	68
53 – 63	69
64 – 74	74
75 – 85	68
86 – 96	24

Exercise 2

Use the frequency distribution in exercise 2 to construct an expended frequency distribution.

Class	Frequency, f	Midpoint	Relative Frequency	Cumulative Frequency
	$\Sigma f =$		$\Sigma \frac{f}{n} =$	

Exercise 3

Construct a frequency distribution and a frequency histogram for the data set using the indicated number of classes. Describe any patterns.

Sales

Number of classes: 6

Data set: July sales (in dollars) for all sales representatives at a company.

2114	2468	7119	1876	4105	3183	1932	1355
4278	1030	2000	1077	5835	1512	1697	2478
3981	1643	1858	1500	4608	1000		

Exercise 4

Construct a frequency distribution and a relative frequency histogram for the data set using five classes. Which class has the greatest relative frequency, and which has the least relative frequency?

Bowling Scores

Data set: Bowling scores of a sample of league numbers

154	257	195	220	182	240	177	228	235
146	174	192	165	207	185	180	264	169
225	239	148	190	182	205	148	188	

Exercise 5

Construct a cumulative frequency distribution and an ogive for the data set using six classes. Then describe the location of the greatest increase in frequency.

Retirement Ages

Data set: Retirement ages for a sample of doctors

70	54	55	71	57	58	63	65
60	66	57	62	63	60	63	60
66	60	67	69	69	52	61	73

Exercise 6

Construct frequency distribution and a frequency polygon for the data set.
Describe any patterns.

Exam Scores

Number of classes: 5

Data set: Exam scores for all students in a statistics class.

83	92	94	82	73	98	78	85	72	90
89	92	96	89	75	85	63	47	75	82